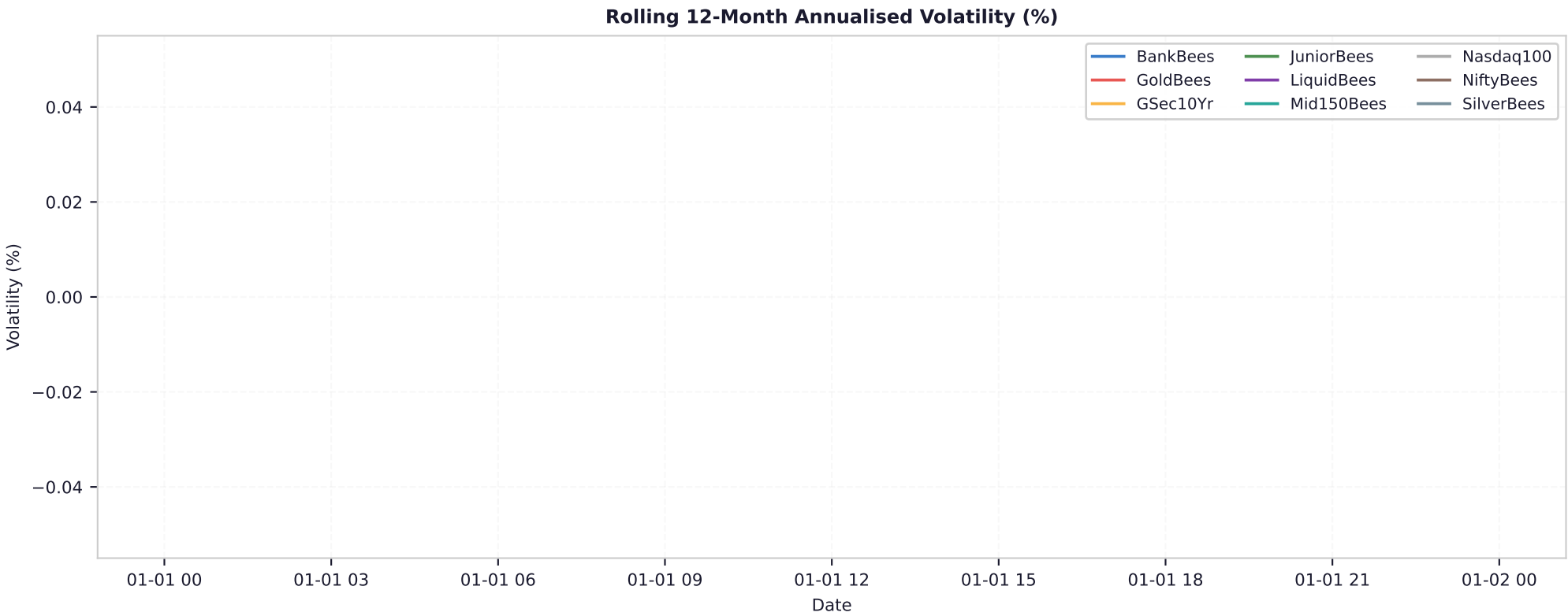
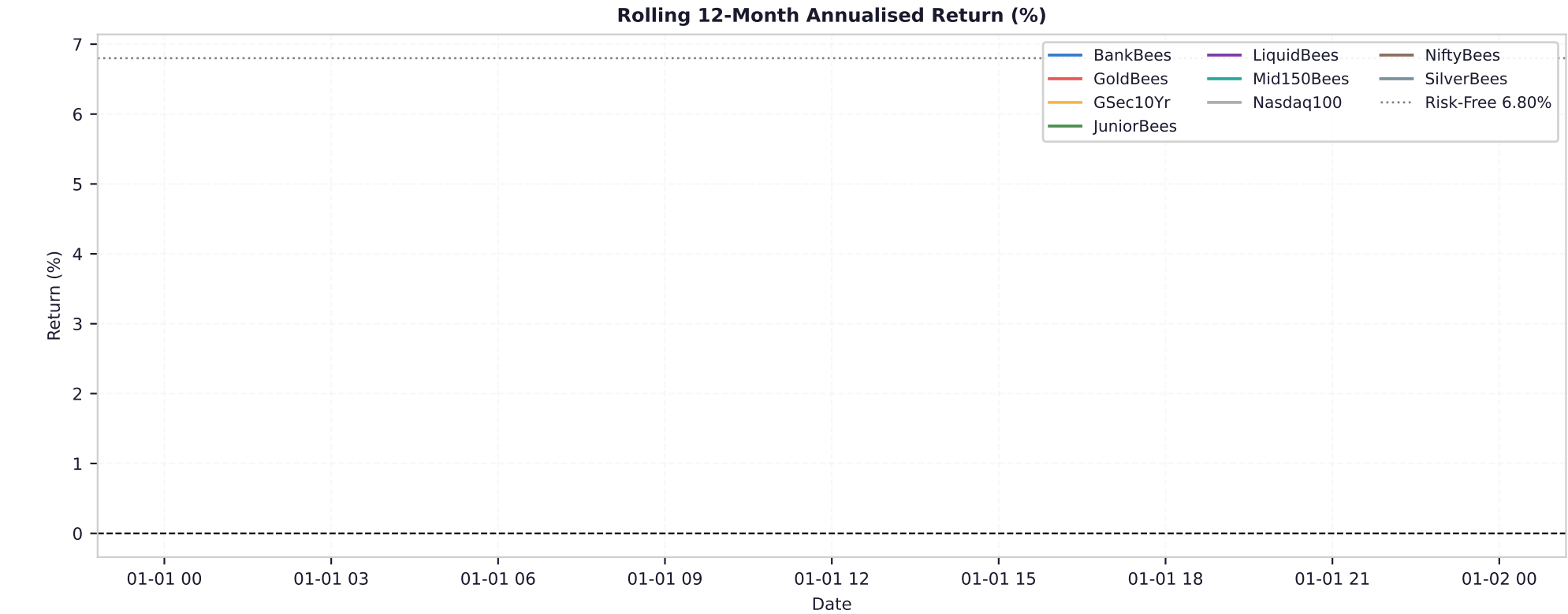
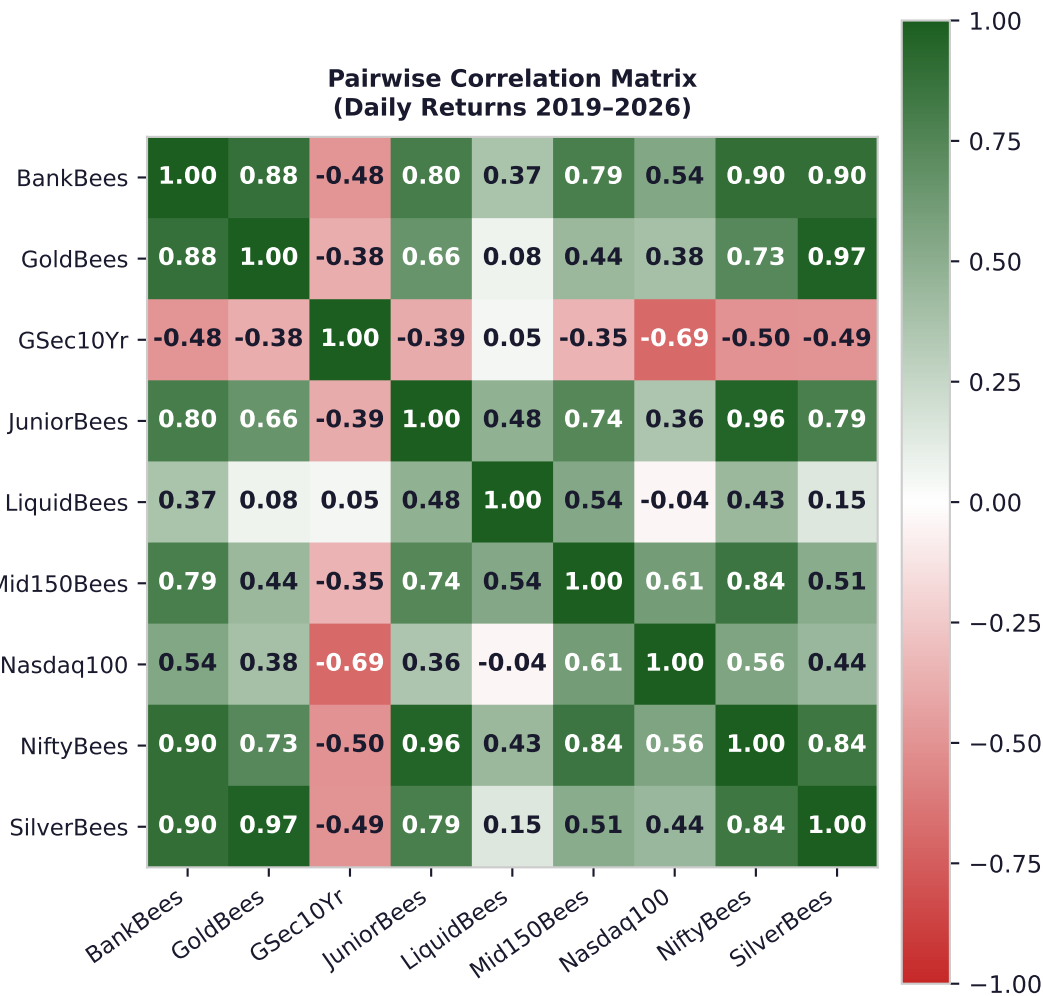


6. Correlation Matrix Analysis

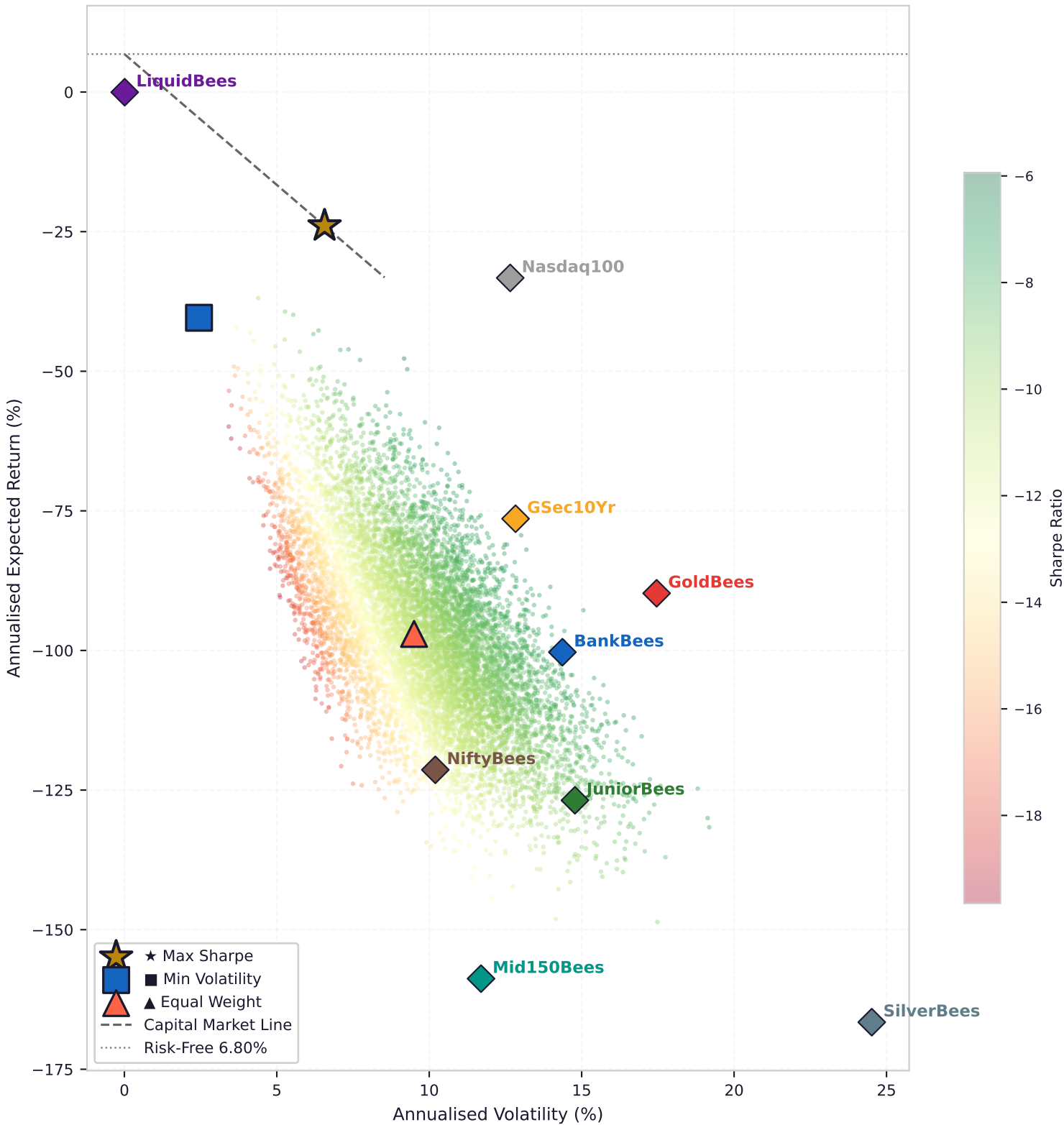


Correlation Insights:

KEY INSIGHTS: NiftyBees and BankBees show very high positive correlation (>0.80), suggesting they move largely together — holding both provides limited diversification benefit. GoldBees is the natural diversifier with near-zero or slightly negative correlation to equity ETFs. LiquidBees is essentially uncorrelated to all other assets, acting as a pure cash proxy. JuniorBees (mid-cap tilt) is highly correlated with NiftyBees but captures the size-premium factor.

7. Efficient Frontier — 10,000 Monte Carlo Portfolios

Efficient Frontier with Capital Market Line



Monte Carlo Stats

Total simulations: 10,000
Max Sharpe found: -5.9422
Min Sharpe found: -19.6508
Median Sharpe: -10.5557
Mean Sharpe: -10.9524

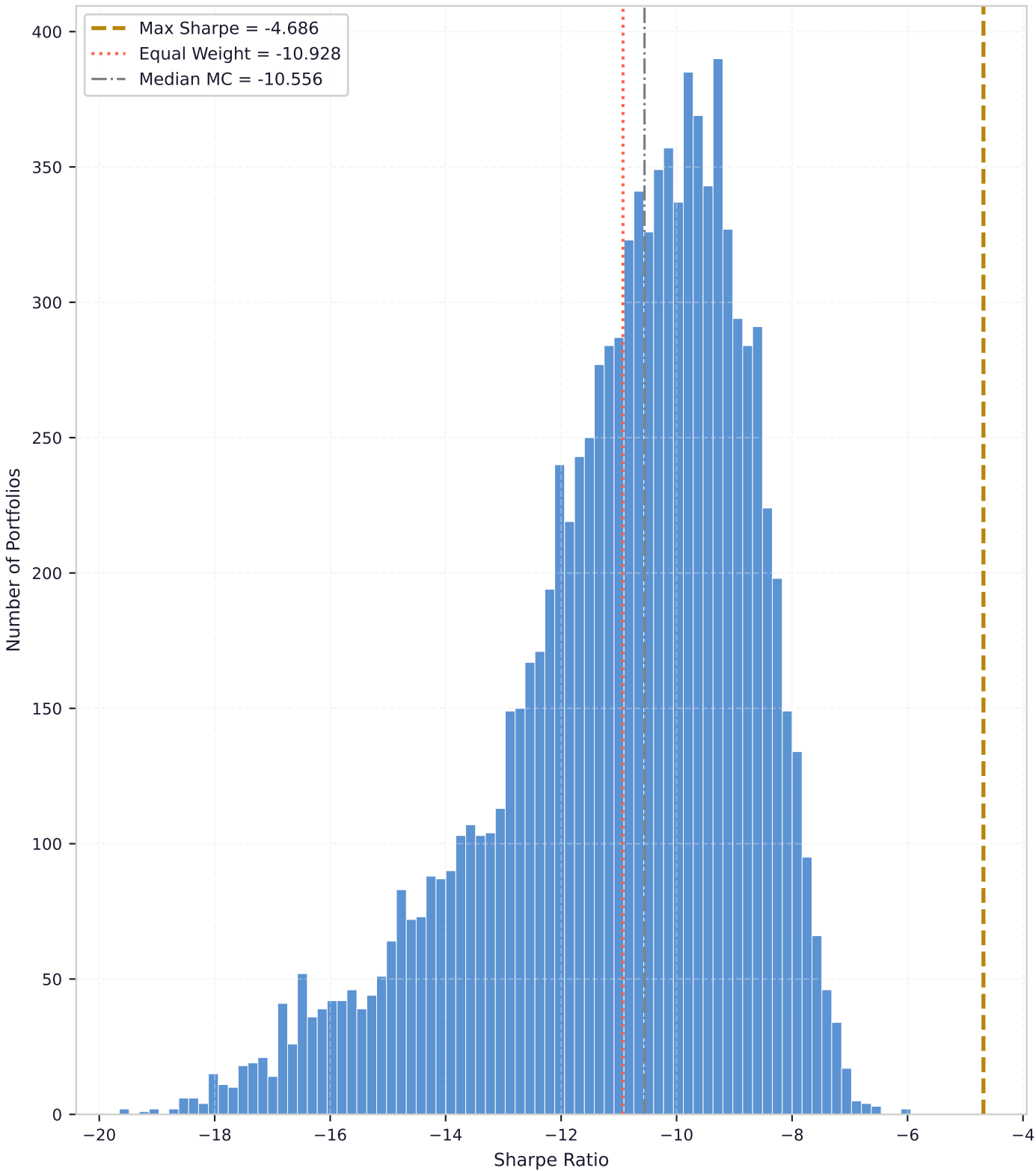
Max Sharpe Portfolio

Return: -23.97% p.a.
Volatility: 6.57% p.a.
Sharpe: -4.6857
WACC: -23.97% p.a.
Status: Converged

Min Volatility Portfolio

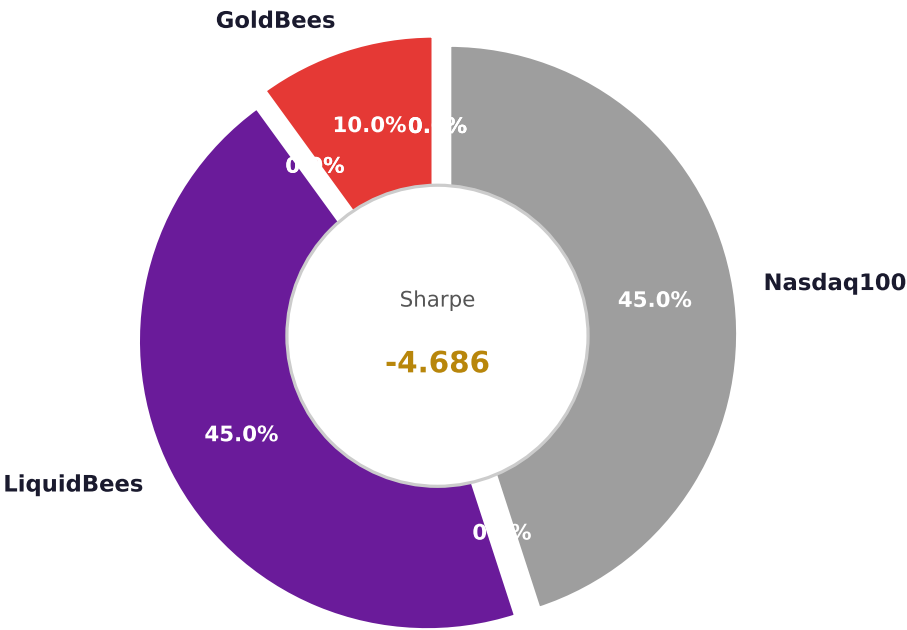
Return: -40.42% p.a.
Volatility: 2.45% p.a.
Sharpe: -19.2923
WACC: -40.42% p.a.
Status: Converged

Distribution of Sharpe Ratios across Monte Carlo Portfolios

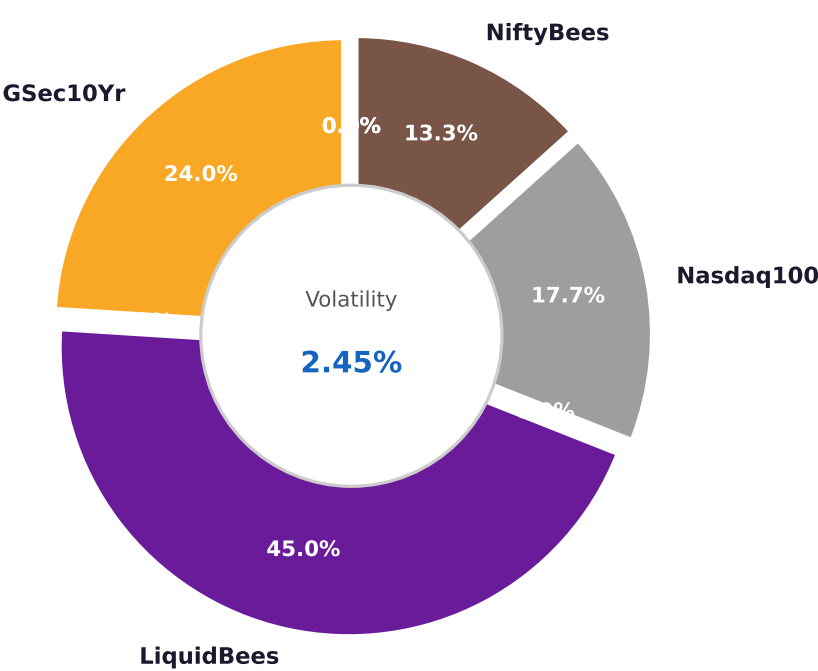


8. Portfolio Allocation Breakdown

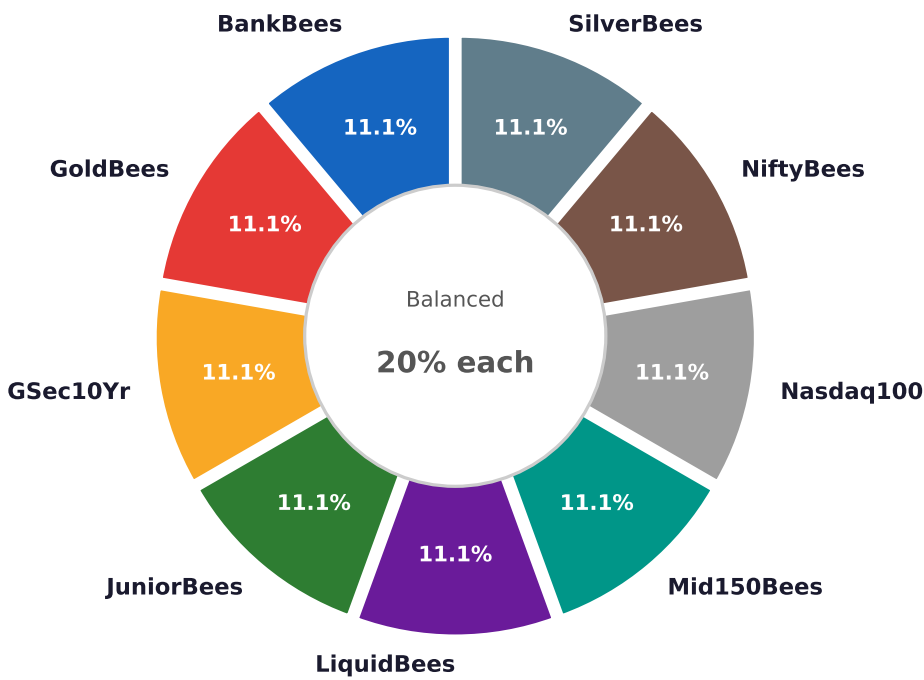
★ Max Sharpe Portfolio



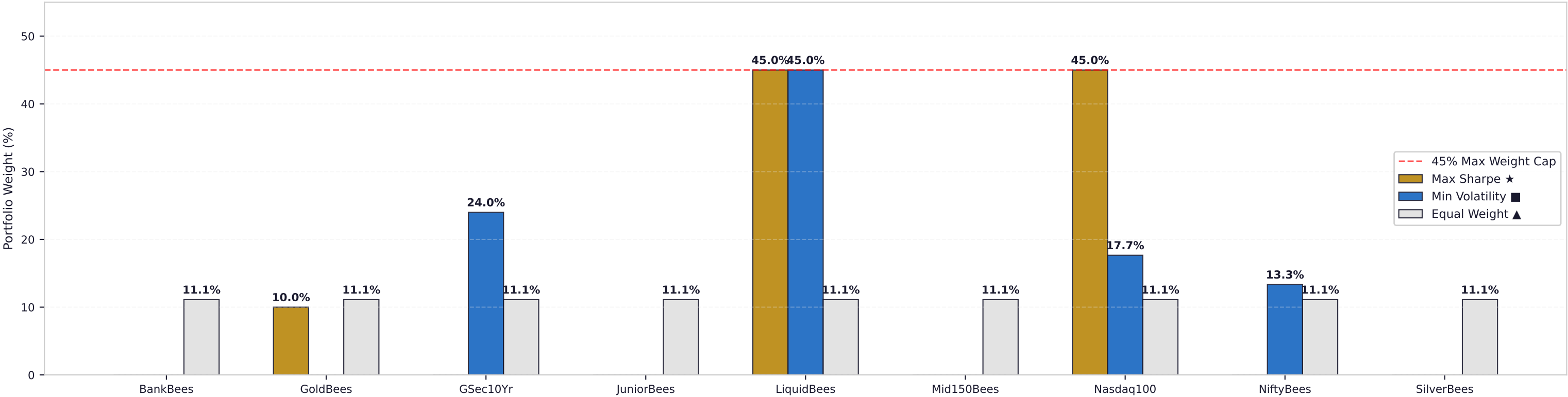
■ Min Volatility Portfolio



▲ Equal Weight (Benchmark)



Weight Comparison Across Three Portfolios



9. Cumulative Growth — ₹100 Invested (2019-2026)

